THE SEPARATION AXIOMS ARE STABLE UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

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Editorial note. The delivered context span is the author's own definition of the separation axioms for a morphism of formal algebraic spaces; the statement depends on it through its reference and it is copied whole.

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Allowed standard prerequisites (editorial).

- The list of separation axioms for morphisms of formal algebraic spaces, each read off from a property of the diagonal. Same chapter, Definition definition-separated-morphism, invoked at proof lines 6822-6823 and 6832-6833. Fixes the list of axioms to be checked and supplies the reading of each of them on the diagonal.
- Stability under base change of the properties of the diagonal of a morphism between formal algebraic spaces. The corresponding stability statements for representability, monomorphism, local quasi-finiteness, local finiteness of type, separatedness and closedness, as recalled in the chapter's separation section. Each property on the list is stable under base change, which transfers it from the diagonal of f to the diagonal of its base change.

Definition 0.1. Let S be a scheme. Let $f : X \to Y$ be a morphism of formal algebraic spaces over S. Let $\Delta_{X/Y} : X \to X \times_Y X$ be the diagonal morphism.

(1) We say f is *separated* if $\Delta_{X/Y}$ is a closed immersion.
(2) We say f is *quasi-separated* if $\Delta_{X/Y}$ is quasi-compact.

Lemma 0.2. *All of the separation axioms listed in Definition 0.1 are stable under base change.*

Proof. Let $f : X \to Y$ and $Y' \to Y$ be morphisms of formal algebraic spaces. Let $f' : X' \to Y'$ be the base change of f by $Y' \to Y$. Then $\Delta_{X'/Y'}$ is the base change of $\Delta_{X/Y}$ by the morphism $X' \times_{Y'} X' \to X \times_Y X$. Each of the properties of the diagonal used in Definition 0.1 is stable under base change. Hence the lemma is true. □

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